Foundations of Uncertainty quantification

[Uncertainty quantification]

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Modular approach An approach to inverse uncertainty quantification is the modular Bayesian approach. The modular Bayesian approach derives its name from its four-module procedure. Apart from the current available data, a prior distribution of unknown parameters should be assigned.

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where y m (x , θ) {\displaystyle y^{m}(\mathbf {x} ,{\boldsymbol {\theta }})} denotes the computer model response that depends on several unknown model parameters θ {\displaystyle {\boldsymbol {\theta }}} , and θ ∗ {\displaystyle {\boldsymbol {\theta }}^{*}} denotes the true values of the unknown parameters in the course of experiments. The objective is to either estimate θ ∗ {\displaystyle {\boldsymbol {\theta }}^{*}} , or to come up with a probability distribution of θ ∗ {\displaystyle {\boldsymbol {\theta }}^{*}} that encompasses the best knowledge of the true parameter values.

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Types of problems There are two major types of problems in uncertainty quantification: one is the forward propagation of uncertainty (where the various sources of uncertainty are propagated through the model to predict the overall uncertainty in the system response) and the other is the inverse assessment of model uncertainty and parameter uncertainty (where the model parameters are calibrated simultaneously using test data). There has been a proliferation of research on the former problem and a majority of uncertainty analysis techniques were developed for it. On the other hand, the latter problem is drawing increasing attention in the engineering design community, since uncertainty quantification of a model and the subsequent predictions of the true system response(s) are of great interest in designing robust systems.

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Uncertainty quantification (UQ) is the science of quantitative characterization and estimation of uncertainties in both computational and real world applications. It tries to determine how likely certain outcomes are if some aspects of the system are not exactly known. An example would be to predict the acceleration of a human body in a head-on crash with another car: even if the speed was exactly known, small differences in the manufacturing of individual cars, how tightly every bolt has been tightened, etc., will lead to different results that can only be predicted in a statistical sense. Many problems in the natural sciences and engineering are also rife with sources of uncertainty. Computer experiments on computer simulations are the most common approach to study problems in uncertainty quantification.

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To evaluate low-order moments of the outputs, i.e. mean and variance. To evaluate the reliability of the outputs. This is especially useful in reliability engineering where outputs of a system are usually closely related to the performance of the system. To assess the complete probability distribution of the outputs. This is useful in the scenario of utility optimization where the complete distribution is used to calculate the utility. To estimate uncertainty in values, that cannot be directly measured, when calculated with experimentally measurable values. This is especially useful in experimental physics and chemistry contexts.

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Frequentist In regression analysis and least squares problems, the standard error of parameter estimates is readily available, which can be expanded into a confidence interval. If the goal is uncertainty quantification for future observations, one may use conformal prediction, which circumvents parameter estimation.

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Simulation-based methods: Monte Carlo simulations, importance sampling, adaptive sampling, etc. General surrogate-based methods: In a non-intrusive approach, a surrogate model is learnt in order to replace the experiment or the simulation with a cheap and fast approximation. Surrogate-based methods can also be employed in a fully Bayesian fashion. This approach has proven particularly powerful when the cost of sampling, e.g. computationally expensive simulations, is prohibitively high. Local expansion-based methods: Taylor series, perturbation method, etc. These methods have advantages when dealing with relatively small input variability and outputs that don't express high nonlinearity. These linear or linearized methods are detailed in the article Uncertainty propagation. Functional expansion-based methods: Neumann expansion, orthogonal or Karhunen–Loeve expansions (KLE), with polynomial chaos expansion (PCE) and wavelet expansions as special cases. Most probable point (MPP)-based methods: first-order reliability method (FORM) and second-order reliability method (SORM). Numerical integration-based methods: Full factorial numerical integration (FFNI) and dimension reduction (DR). For non-probabilistic approaches, interval analysis, Fuzzy theory, Possibility theory and evidence theory are among the most widely used. The probabilistic approach is considered as the most rigorous approach to uncertainty analysis in engineering design due to its consistency with the theory of decision analysis. Its cornerstone is the calculation of probability density functions for sampling statistics. This can be performed rigorously for random variables that are obtainable as transformations of Gaussian variables, leading to exact confidence intervals.

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The opinion triangles represent barycentric coordinate systems, where a subjective opinion is visualized as a blue dot, with a blue frame for associated values for Belief, Disbelief and E-uncertainty (epistemic), where Belief + Disbelief + E-uncertainty = 1. The blue frame also shows base rate (prior) and probability values. A-uncertainty (aleatoric) is greatest when the probability is P = 0.5. The Beta probability density function to the right of the triangles visualizes where on the scale [0, 1] the probability has the greatest density, i.e. where the probability presumably lies. See the subjective opinion online demo of uncertainty visualization.

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For most cases, p (θ , ϕ ■ data) {\displaystyle p({\boldsymbol {\theta }},{\boldsymbol {\varphi }}\mid {\text{data}})} is a highly intractable function of ϕ {\displaystyle {\boldsymbol {\varphi }}} . Hence the integration becomes very troublesome. Moreover, if priors for the other hyperparameters ϕ {\displaystyle {\boldsymbol {\varphi }}} are not carefully chosen, the complexity in numerical integration increases even more. In the prediction stage, the prediction (which should at least include the expected value of system responses) also requires numerical integration. Markov chain Monte Carlo (MCMC) is often used for integration; however it is computationally expensive. The fully Bayesian approach requires a huge amount of calculations and may not yet be practical for dealing with the most complicated modelling situations.

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Uncertainty propagation is the quantification of uncertainties in system output(s) propagated from uncertain inputs. It focuses on the influence on the outputs from the parametric variability listed in the sources of uncertainty. The targets of uncertainty propagation analysis can be: